



Computer Graphics 2 - Exercise Sheet 2

Technische Universität Berlin - Computer Graphics

Handout 14. May 2026, **Practice Due** 03. June 2026, **Theory Due** 03. June 2026 11:59 pm

Hendrik Meyer, Gabriel Dogadov, Prof. Dr. Marc Alexa

Exercise 1: Approximation of height fields (5 points)

The objective of this exercise is to approximate a parametric surface from a set of points (x_i, y_i, z_i) . The surface is modeled as $\mathbf{q}(x, y) = (x, y, z(x, y))$ with the x - y coordinates lying on a regular grid, and can be visualized with a triangle or quad mesh. The framework developed in the first exercise should be used to efficiently query points. The implementation includes the following tasks (one point per task):

1. Load data points (x_i, y_i, z_i) from an OFF file into your spatial data structure and compute their minimum bounding box. Subsequently, generate a regular $m \times n$ grid in the x - y domain of the bounding box, with adjustable parameters m and n . Approximate the z -coordinate of the grid points by weighted least squares (WLS) from the input points using a suitable weighting function with local support (e.g., Wendland's function), where f is a bivariate quadratic polynomial. Note that all distances for the weighting function are computed in the x - y domain.
2. We now want to create a Bézier tensor product surface using the grid points as control points. Therefore, evaluate $km \times kn$ points on the surface by using de Casteljau's algorithm. Visualize the surface as a surface mesh.
3. Use de Casteljau's algorithm to *efficiently* estimate the surface normals of the Bézier tensor product surface and visualize them.
4. Again, create a regular grid in the x - y domain, but this time with a higher resolution of $km \times kn$, $k \in \mathbb{N}^+$ being an adjustable parameter. Compute the z -coordinate of the grid points using WLS. Visualize the resulting MLS surface as a polygon mesh.
5. Approximate the normals of the MLS surface (see theory task 5) and visualize them.

Exercise 2: Theory (6 + 2 points)

1. Derive the basis functions of a first order (= degree 1) B-Spline with knot vector $(0, 0, 1, 2, 3, 4, 4)$. Then, draw the first order B-spline for the control points $(0,0)$, $(1,3)$, $(2,1)$, $(2,3)$ and $(4,2)$. Explain the **connection** between degree, continuity, and support of B-Splines at this example. (2 points)
2. Let a curve be defined as the affine combination of control points $\mathbf{p}_i$, where the weights are given by a set of basis functions. Show that the curve is invariant under affine transformations if the basis functions are a partition of unity in the interval defining the curve. (1 point)
3. There are multiple algorithm variations of de Casteljau's algorithm possible for the evaluation of a Bézier tensor product surface. The two presented in the lecture are (1) row-first evaluation using the 1D de Casteljau algorithm and (2) column-first evaluation using the 1D de Casteljau's algorithm. Compare the number of elementary arithmetic operations (i.e., addition and multiplication) needed to evaluate a single point $\mathbf{q}(u, v)$ of the surface $\mathbf{q}$ with respect to the number of column control points n_c and row control points n_r . (1 point)
4. Is the tensor product of two linear polynomials a flat surface? Explain your answer. (1 point)
5. As a simple approximation, the tangents of a local polynomial approximation can be used as the tangents of the MLS surface (computed from the continuously moving weighted least squares approximation). Show that the tangents of the local polynomial are not necessarily the tangents of the MLS surface. (1 point)
Derive an exact equation for the tangents. (2 bonus points)